

Bank Customer Churn Analysis

1. Project Overview

This project analyzes customer churn behavior for a retail bank using a dataset of 10,000 customer records. The goal is to identify the demographic, financial, and engagement factors that drive customers to leave the bank, quantify the revenue impact of churn, and provide actionable recommendations for retention strategy.

2. Dataset Summary

- Rows: 10,000
- Columns: 13 (original), 17 (after feature engineering)
- Key Features:
 - Customer demographics (CustomerId, Surname, Geography, Gender, Age)
 - Account details (CreditScore, Tenure, Balance, NumOfProducts, HasCrCard, EstimatedSalary)
 - Engagement (IsActiveMember)
 - Target variable (Exited – 1 if the customer churned, 0 if retained)
 - Missing Data: none (0 nulls across all 13 original columns)

3. Exploratory Data Analysis using Python

Data preparation and cleaning were carried out in Python:

- Data Loading: Imported the dataset (Bank_Churn.csv) using `pandas`.
- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

[4]:

	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary
count	1.000000e+04	10000	10000.000000	10000	10000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
unique	NaN	2932	NaN	3	2	NaN	NaN	NaN	NaN	NaN	NaN	NaN
top	NaN	Smith	NaN	France	Male	NaN	NaN	NaN	NaN	NaN	NaN	NaN
freq	NaN	32	NaN	5014	5457	NaN	NaN	NaN	NaN	NaN	NaN	NaN
mean	1.569094e+07	NaN	650.528800	NaN	NaN	38.921800	5.012800	76485.889288	1.530200	0.70550	0.515100	10000.000000
std	7.193619e+04	NaN	96.653299	NaN	NaN	10.487806	2.892174	62397.405202	0.581654	0.45584	0.499797	575.000000
min	1.556570e+07	NaN	350.000000	NaN	NaN	18.000000	0.000000	0.000000	1.000000	0.000000	0.000000	0.000000
25%	1.562853e+07	NaN	584.000000	NaN	NaN	32.000000	3.000000	0.000000	1.000000	0.000000	0.000000	5100.000000
50%	1.569074e+07	NaN	652.000000	NaN	NaN	37.000000	5.000000	97198.540000	1.000000	1.000000	1.000000	10000.000000
75%	1.575323e+07	NaN	718.000000	NaN	NaN	44.000000	7.000000	127644.240000	2.000000	1.000000	1.000000	14930.000000
max	1.581569e+07	NaN	850.000000	NaN	NaN	92.000000	10.000000	250898.090000	4.000000	1.000000	1.000000	19990.000000

EstimatedSalary	Exited
10000.000000	10000.000000
NaN	NaN
NaN	NaN
NaN	NaN
100090.239881	0.203700
57510.492818	0.402769
11.580000	0.000000
51002.110000	0.000000
100193.915000	0.000000
149388.247500	0.000000
199992.480000	1.000000

- **Missing Data Handling:** Confirmed via `df.isnull().sum()` that there were no missing values in the raw dataset.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by customer ages into 18-30, 31-40, 41-50, 51-60, and 60+.
 - Created **balance_segment** column by binning account balance into Zero, Low (0-75K), Medium (75K-150K), and High (150K+).
 - Created **credit_tier** column by binning **credit_score** into Poor, Fair, Good, and Excellent.
 - Created **tenure_segment** column by binning tenure into New (0-2yr), Regular (3-6yr), and Loyal (7+yr).
- **Data Consistency Check:** Confirmed an overall churn rate of 20.37% (2,037 of 10,000 customers exited).
- **Database Integration:** Connected the Python script to a local SQL Server and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

4.1. Churner vs. Retained Profile

Compared average age, credit score, balance, tenure, and activity rate between customers who churned and those who stayed.

	customer_status	total	avg_age	avg_credit_score	avg_balance	avg_tenure_yrs	pct_active
1	Retained	7963	37.4	652	72745	5	55.5
2	Churned	2037	44.8	645	91109	4.9	36.1

4.2. Highest Churn Risk Segment (Geography, Gender, Age Group)

Identified the combination of geography, gender, and age group with the highest churn rate (segments with at least 30 customers).

	geography	gender	age_group	total_customers	total_churned	churn_rate_pct
1	Germany	Female	51-60	129	95	0.736434108527132
2	Germany	Male	51-60	114	74	0.649122807017544
3	France	Female	51-60	171	110	0.64327485380117
4	Spain	Female	51-60	91	50	0.549450549450549
5	Germany	Female	41-50	330	173	0.524242424242424
6	Germany	Female	60+	53	27	0.509433962264151
7	Germany	Male	41-50	348	158	0.454022988505747
8	France	Male	51-60	196	83	0.423469387755102
9	Spain	Male	51-60	96	36	0.375
10	Spain	Female	41-50	271	91	0.335793357933579
11	France	Female	41-50	499	164	0.328657314629259
12	Germany	Male	60+	60	17	0.283333333333333
13	Germany	Female	31-40	473	120	0.25369978858351
14	France	Female	60+	111	28	0.252252252252252
15	France	Male	41-50	568	136	0.23943661971831
16	Spain	Male	60+	71	16	0.225352112676056
17	Spain	Male	41-50	304	66	0.217105263157895
18	Spain	Female	60+	49	9	0.183673469387755
19	Germany	Male	31-40	549	94	0.171220400728597
20	Germany	Female	18-30	208	33	0.158653846153846
21	France	Male	60+	120	18	0.15
22	France	Female	31-40	1018	130	0.12770137524558
23	Spain	Female	18-30	210	26	0.123809523809524
24	Spain	Female	31-40	468	55	0.117521367521368
25	Germany	Male	18-30	245	23	0.0938775510204...
26	Spain	Male	31-40	650	49	0.0753846153846...
27	France	Male	31-40	1293	90	0.0696055684454...
28	France	Female	18-30	462	28	0.0606060606060...
29	Spain	Male	18-30	267	15	0.0561797752808...
30	France	Male	18-30	576	23	0.0399305555555...

4.3. Number of Products vs. Churn

Examined whether holding more bank products increases loyalty or drives churn.

	num_of_products	total_customers	total_churned	churn_rate_pct
1	1	5084	1409	27.700000000000
2	2	4590	348	7.600000000000
3	3	266	220	82.700000000000
4	4	60	60	100.000000000000

4.4. Engagement and Balance Impact on Churn

Analyzed how account activity (active vs. inactive) combined with balance tier (Zero, Mid up to 100K, High above 100K) impacts churn likelihood.

	engagement	balance_tier	total_customers	churned	churn_rate_pct
1	Inactive	High	2356	772	32.800000000000
2	Inactive	Mid	749	210	28.000000000000
3	Inactive	Zero	1744	320	18.300000000000
4	Active	High	2443	439	18.000000000000
5	Active	Mid	835	116	13.900000000000
6	Active	Zero	1873	180	9.600000000000

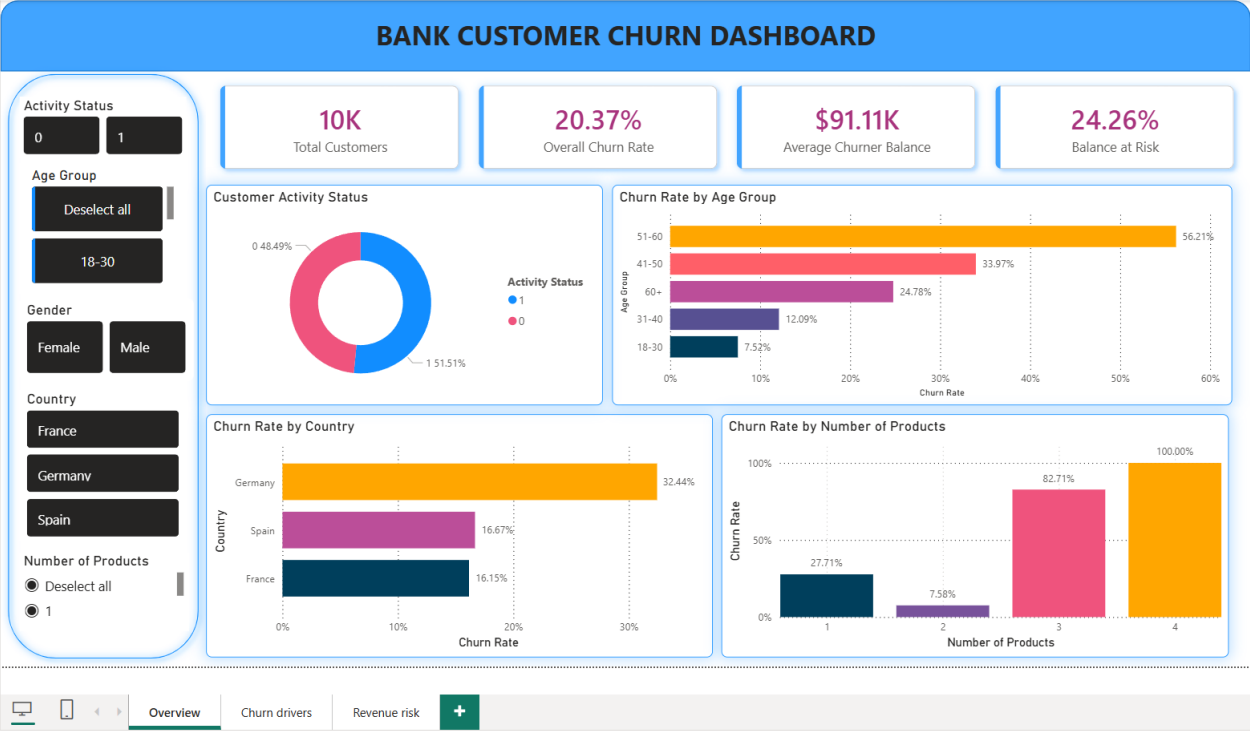
4.5. Financial Impact of Churn by Region

Quantified the total account balance lost due to churn, broken down by geography.

	geography	customers_lost	balance_lost	pct_of_total_balance
1	Germany	814	97973916	12.8
2	France	810	57666165	7.5
3	Spain	413	29948015	3.9

5. Dashboard in Power BI

An interactive dashboard was built in Power BI to present insights visually.





6. Business Recommendations

- **Re-engage Inactive Customers** – Inactive members churn at roughly double the rate of active members across all balance tiers; launch targeted re-engagement campaigns (app reminders, loyalty offers) for inactive accounts, especially those with high balances.
- **Review the 3- and 4-Product Bundles** – Customers holding 3+ products churn at 82.7% to 100%, far above the 7.6% rate for 2-product holders; audit these bundles for fit, pricing, or cross-sell pressure that may be driving dissatisfaction.
- **Prioritize Retention in Germany** – Germany has the highest churn rate (32.4%) and the largest balance loss (~\$98M), particularly among older female customers; deploy region-specific retention offers and relationship-manager outreach.
- **Target Older Customer Segments** – Churn rises sharply with age, peaking at 56.2% for the 51-60 group; design retirement-stage products and proactive check-ins for customers aged 50+.
- **Monitor High-Balance Accounts** – Churned customers carry higher average balances (\$91.1K) than retained customers (\$73K); treat high-balance, low-activity accounts as early-warning churn signals and assign them to retention teams.